Reproducibility Report: Graham and Coppock (2021)

Asking About Attitude Change

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This repository holds the actively maintained replication code for Graham and Coppock (2021), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1093/poq/nfab009
Replication archive	10.7910/DVN/GFF78K
Mirror deposit	10.60600/YU/2ADPVK

Pre-analysis plans	none

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the Dataverse file identifiers, the UNF of each ingested data file, and two checksums per file: the MD5 of the bytes Dataverse serves, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all 12 files. They do not always, so the script verifies against the served bytes and reports any disagreement. It also checks byte sizes and stops if `original/` holds anything the manifest does not list, because an archive script that writes into its own directory adds files as well as overwriting them. Either way the exact bytes are pinned in version control even though the bytes themselves are not. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `graham_coppock_2021.Rproj`, and run:

```r
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies its 12 checksums, produces every published figure, appendix table and in-text number into `maintained/output/`, rebuilds the ground truth from those outputs, and re-checks the deposit. It takes about 75 seconds, of which roughly 65 are the 10,000-resample bootstrap behind Table D.5. Individual scripts run on their own in any order, subject to the dependencies in each script's header: `clean_studies.R` first, then the two estimation scripts, then the figures, tables and in-text scripts that read them.

Required packages: tidyverse, estimatr, ggh4x, gridExtra, rsample, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV and PNG output should come back byte-identical. The six PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Paper Overview

Citation: Graham, M. H. and Coppock, A. (2021). Asking about attitude change. *Public Opinion Quarterly*, 85(1), 28–53.

Research question: When a survey asks people whether some event or piece of information made them more or less supportive of something, do their answers measure attitude change? The paper's claim is that they largely do not: respondents use the change question to express the *level* of their attitude, a bias the paper calls response substitution.

Design: Four survey experiments plus one illustrative survey. The illustration is a 4,034-respondent Lucid survey on impeaching Donald Trump, asked in the paper's proposed *counterfactual format*: respondents report their attitude, then report what they would have said had they not known about the Ukraine affair. Studies 1 (N = 417), 2a (N = 2,475) and 2b (N = 1,110) randomly assign respondents both to an information treatment and to a question format, so the self-reported change can be set against an experimental difference in means on the same treatment. Study 3 (N = 1,074) varies only the format. Estimation is by difference in means and by group means with HC2 standard errors, clustered by respondent where a respondent answers about more than one treatment.

Main finding: The change format performs badly and the counterfactual format performs well. Across ten treatments split by party, the more-minus-less statistic derived from the change format has the opposite sign to the experimental benchmark in 12 of 20 comparisons, against 3 of 20 for the counterfactual format. Asking a level question first cuts reports of any change by about 10 percentage points, which is what response substitution predicts.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Not as deposited. `analysis_impeach.R`, which draws Figure 1, runs to completion. `analysis_main.R`, which produces everything else, fails twice.

- It fails first at `ggsave("drafts/POQ_RR/figures/substitution_new.png", ...)`, one of six active `ggsave()` calls writing into two directory trees on the authors' own machine that the deposit does not contain. Every analysis in the script completes; only the writes fail.

- With those directories created it fails again, and this one is substantive: `object 'check_labeller' not found`. The archive ships its own `facet_nested()` in `functions.R`, copied from an early standalone script, and that function calls `ggplot2:::check_labeller()` and `ggplot2:::var_list()`. Both are unexported internals and both have since been removed from ggplot2. The function is used for every panel of Figures 2, 4, 5 and E.2, so the failure takes out four of the paper's six figures.

Neither failure needed the authors' help or a file the archive omits. Substituting `ggh4x::facet_nested()`, the maintained successor to the same code, is enough to make the whole script run.

Two smaller things are worth recording because a reader reproducing this work will meet them. `analysis_impeach.R` writes `GrahamCoppock19-0280-Figure1.jpeg` to a bare relative path, and both scripts leave an `Rplots.pdf` behind when run non-interactively, so running the archive in its own directory silently adds two files to the deposit. And `functions.R` calls `plyr` in eight places without any script declaring it, while `analysis_main.R` loads `ggpubr`, `janitor` and `ggrepel` and never calls a function from any of them.

2.2 Does the maintained rewrite reproduce the paper?

Almost entirely. Every appendix table reproduces cell for cell, and 50 of the 57 recorded claims match.

Table 1: Reproduction verdict by component

Component	Verdict
Figure 1b (seven printed group means)	All seven reproduce
Figure 2 (ten printed labels)	All ten reproduce
Figure 3 (shape and the effects the text quotes)	Reproduces
Figure 4 (96 printed labels)	95 of 96 reproduce; the miss is a rounding tie
Figure 5 (24 printed labels)	All 24 reproduce
Figure E.2 (44 printed labels)	All 44 reproduce
Table D.3 (336 cells)	All reproduce, with the published column headings transposed
Table D.4 (216 cells, 72 p-values)	All reproduce
Table D.5 (60 point estimates)	All reproduce
Table D.5 (bootstrap intervals)	53 of 60 rows; the deposited script reproduces all 60
In-text numbers	All but two reproduce; both are the article rounding its own estimate

The exceptions are small and none of them is a failure of the analysis.

The article rounds two of its own numbers away from what its data give. The text reports the two impeachment questions correlating at 0.82; the deposited data give 0.828, which prints as 0.83. It reports the level question reducing Democrats' reports of change by 14 points; the estimate is 13.5 points, which prints as 13. Both are one unit in the last printed digit, and neither touches a conclusion.

Study 3's published N cannot be recovered from the deposit. The article reports 1,074 respondents. The deposited data hold 1,023 distinct Study 3 respondents, in the standalone `data_study3.csv` and in the merged file alike. The published figure is the full Mechanical Turk survey and the deposit carries only the respondents in a question format condition; Table D.3's Study 3 rows reproduce exactly from what is there.

Table D.3's column headings are transposed relative to its own body. The published table heads its second and third value columns "No change" and "Less", but in all 84 rows the printed Difference equals More minus the column headed "No change". Read in that corrected order all 336 cells reproduce. Read as printed, 172 do. The numbers in the table are right; the two headings are swapped.

One printed label in Figure 4 differs, and it is a rounding tie. For the Undisputed accusation treatment among Republicans, the counterfactual-format estimate is a mean that comes to exactly -1.125. `lm_robust` returns it one unit in the last place away from that tie, so it prints as -1.13, which is what both the deposited script and this rewrite produce. The published figure prints -1.12, which is what the tie itself gives, and which is also what the appendix's Table D.5 prints for the same quantity. Nothing about the estimate differs; only the last printed digit does.

Table D.5's bootstrap intervals are the one place the rewrite and the deposit part company. Both draw 10,000 resamples from seed 0. The deposited script resamples every cell in the counterfactual data and then keeps the twenty the table reports; the rewrite selects the twenty first, so the two walk different random streams from the same seed. The deposited script reproduces all sixty published rows exactly. The rewrite differs on seven rows by a unit in the second decimal, which is Monte Carlo variation rather than a difference in the estimator, and no interval changes whether it covers zero.

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3 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/GFF78K, 12 files.

The archive contains two analysis scripts, a shared `functions.R`, a codebook, a readme, and seven data files: the merged Studies 1 to 3 frame and the impeachment survey as `.rds`, and both plus the four individual studies as `.csv`.

Table 2: Original archive reproducibility check (current R environment)

Script	Status	Issue
functions.R	Sourced by both	Custom facet_nested() calls two ggplot2 internals that no longer exist; undeclared dependency on plyr
analysis_impeach.R	Runs	Writes a JPEG to a bare relative path, and an Rplots.pdf under Rscript
analysis_main.R	FAILS	Six ggsave() calls to directories the deposit does not contain, then facet_nested()

Failure details.

- `analysis_main.R` lines 192, 193, 531, 545, 551 and 566 call `ggsave()` into `drafts/POQ_RR/figures/` and `drafts/POQ_final/final figures/`. Neither directory is in the deposit. The script's own comments elsewhere say to change the paths and uncomment the calls, and most of the `ggsave()` calls are indeed commented out; these six are not.
- `functions.R` defines a 250-line `facet_nested()` that predates the version now in `ggh4x`. It calls `ggplot2:::check_labeller()` at line 159 and `ggplot2:::var_list()` at line 335. Both were unexported and both are gone from current ggplot2, and either is fatal. `ggh4x::facet_nested()` is the maintained successor and is a drop-in replacement here.
- `functions.R` also defines a `position_jitter_ellipse()` ggproto that calls `ggplot2:::with_seed_null()`. That internal still exists, so Figure 1 draws today. It is an unexported function nonetheless, and the rewrite uses the standard `position_jitter()` instead; the panel carries no numbers and the article describes the jitter only as "a small amount random noise added to distinguish the points."

Deprecations that warn rather than fail: `%>%`, `do(tidy(...))`, `gather()` and `spread()`, `geom_errorbar()` for confidence bars, `size =` for line widths, `ifelse()`, `rm(list = ls())` at the top of both scripts, and `source("functions.R")` by relative path.

One coding slip, carried forward rather than corrected. Assembling the estimates for Figures 2, 4, 5 and E.2, the deposited script sorts the topics with `factor(Study, c(1, 4, 2, 3))`. The study labels are `1`, `2a`, `2b` and `3`, so `2a` and `2b` do not match any level and become `NA`, sorting last. That ordering is what produced the published facet sequence, so the rewrite keeps it; it uses the sort key and then drops it, rather than carrying a study column that is empty for half the rows.

4 Ground Truth

`ground_truth/build_ground_truth.R` writes `graham_coppock_2021_ground_truth.csv` and runs as the last step of `run_all.R`, so the table cannot drift from the pipeline. Published values are read from the article and the supplementary materials and live either in that script or in the four `published_*.csv` files beside it, which transcribe Tables D.3, D.4 and D.5 and the printed labels of Figures 2, 4, 5 and E.2 in the layout the pages print them in. Every value in the `value_rewrite` column is read back out of `maintained/output/`; none is typed. The verdicts are computed, by printing the rewrite's number to the precision the page prints and comparing digits.

`ground_truth/extract_archive_values.R` fills the `value_script` column. It runs the deposited scripts in a scratch copy, with the two repairs above, and compares each of their objects against the maintained output it corresponds to.

Table 3: Deposited scripts against the maintained rewrite, object by object

Object	Archive rows	Rewrite rows	Joined	Largest difference
grand comparison estimates	480	480	480	0.0000
pretreated estimates	428	428	428	0.0000
Figure 1b group means	7	7	7	0.0000
Figure 3a estimates	30	40	30	0.0000
Table D.3 cells	336	336	336	0.0000
Table D.4 cells	216	216	216	0.0000
Table D.5 point estimates	60	60	60	0.0000
Table D.5 bootstrap quantities	60	60	60	0.0109

Everything except the Table D.5 bootstrap agrees to machine precision. The Figure 3a row joins 30 of the rewrite's 40 estimates because the rewrite also estimates the pure independents that the panel leaves out and the surrounding text quotes.

Table 4: Ground truth: 50 of 57 claims match

Where the difference lies	Rows
matches	50
paper_internal	3
environment	2
archive	1

rewrite	1

Table 5: The 7 claims that do not match

Claim	Published	Rewrite	Locus
Text, p. 32: Correlation of the two impeachment questions	0.82	0.8285	paper_internal
Text, p. 44: Study 3 N	1074.00	1023.0000	archive
Text, p. 44: Same decrease among Democrats (points)	14.00	13.4590	paper_internal
Figure 4: Printed labels agreeing with the published figure	96.00	95.0000	environment
Figure 4: Undisputed accusation, Republican, CATE, Counterfactual	-1.12	-1.1250	environment
Table D.3: Cells agreeing if the column headings are read as printed	336.00	172.0000	paper_internal
Table D.5: Bootstrap standard errors and interval endpoints agreeing	60.00	53.0000	rewrite

The full table, all 57 rows with their notes, is `ground_truth/graham_coppock_2021_ground_truth.csv`.

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5 Maintained Rewrite

Table 6: Maintained rewrite scripts

Script	Output
helpers.R	Packages, colours, theme, the shared grand comparison panel
clean_studies.R	output/data_study1-3_clean.rds, output/data_impeach_clean.rds
estimates_grand_comparison.R	output/estimates_grand_comparison.{rds,csv}
estimates_pretreated.R	output/estimates_pretreated.{rds,csv}
figure_1_impeach_counterfactual.R	output/figure_1_impeach_counterfactual.{pdf,png,csv}

figure_2_cornish_example.R	output/figure_2_cornish_example.{pdf,png,csv}
figure_3_response_substitution.R	output/figure_3_response_substitution.{pdf,png}, three CSVs
figure_4_grand_comparison.R	output/figure_4_grand_comparison.{pdf,png,csv}
figure_5_pretreat_comparison.R	output/figure_5_pretreat_comparison.{pdf,png,csv}
figure_e2_study2b_simultaneous.R	output/figure_e2_study2b_simultaneous.{pdf,png,csv}
table_d3_change_distribution.R	output/table_d3_change_distribution{,_estimates}.csv
table_d4_counterfactual_accuracy.R	output/table_d4_counterfactual_accuracy{,_estimates}.csv
table_d5_ate_vs_self.R	output/table_d5_ate_vs_self{,_estimates}.csv
text_intext_stats.R	output/text_intext_stats.csv
text_sign_comparison.R	output/text_sign_comparison.csv

Two structural choices are worth naming. Figures 2, 4, 5 and E.2 are the same panel drawn over different subsets, which is how the deposit built them too, so the panel is one function in `helpers.R` and every figure script calls it. And the estimates behind those four figures are computed once, in `estimates_grand_comparison.R` and `estimates_pretreated.R`, so no estimate in the paper is computed twice by two scripts that could then disagree.

Every figure script writes a CSV of what it plots, and every table script writes its cells twice: once formatted as the appendix lays them out, once unrounded. The unrounded copy is what the ground truth reads, so a comparison against a published two-decimal table never has to parse a three-decimal string and round it a second time.

Table 7: Deprecated patterns and replacements

Original	Rewrite
custom facet_nested() using ggplot2 internals	ggh4x::facet_nested()
coord_flip() with Format on the discrete axis	native horizontal layout: y = Format, x = estimate
position_jitter_ellipse() ggproto	position_jitter() with a seed
%>%	|>
do(tidy(...))	reframe(tidy(...))
gather() / spread()	pivot_longer() / pivot_wider()
geom_errorbar(width = 0)	geom_linerange()
ifelse()	if_else()

recode()	recode_values()
replicate(10000, oneDifference(.))	rsample::bootstraps(times = 10000)
xtable + print.xtable to the console	write_csv() to output/
size = for line widths	linewidth =
rm(list = ls())	removed
ggsave() to the authors' own directories	ggsave() through here::here() into maintained/output/
library() in each script	source(helpers.R); all packages in helpers.R

The layout change is the largest one. The deposit draws the grand comparison panel with `coord_flip()` over `facet_nested(Party + Estimator ~ Topic, switch = "y")`. Combined with `space = "free_y"`, that arrangement errors out under `ggh4x`. The rewrite maps the panel natively instead, with `Format` on the vertical axis and the estimate on the horizontal, and facets on `Topic ~ Party + Estimator` with `switch = "x"`. Nothing about the estimates changes and the figures carry the same content in the same arrangement.

The bootstrap is the same estimator by a different engine. The deposit resamples rows by hand inside `replicate()`; the rewrite uses `rsample::bootstraps()` at the same seed and the same 10,000 draws. Both are the nonparametric bootstrap over the same rows, and the point estimates are identical. The intervals differ on seven of sixty rows in the second decimal, for the reason given in the Summary.

6 Figures

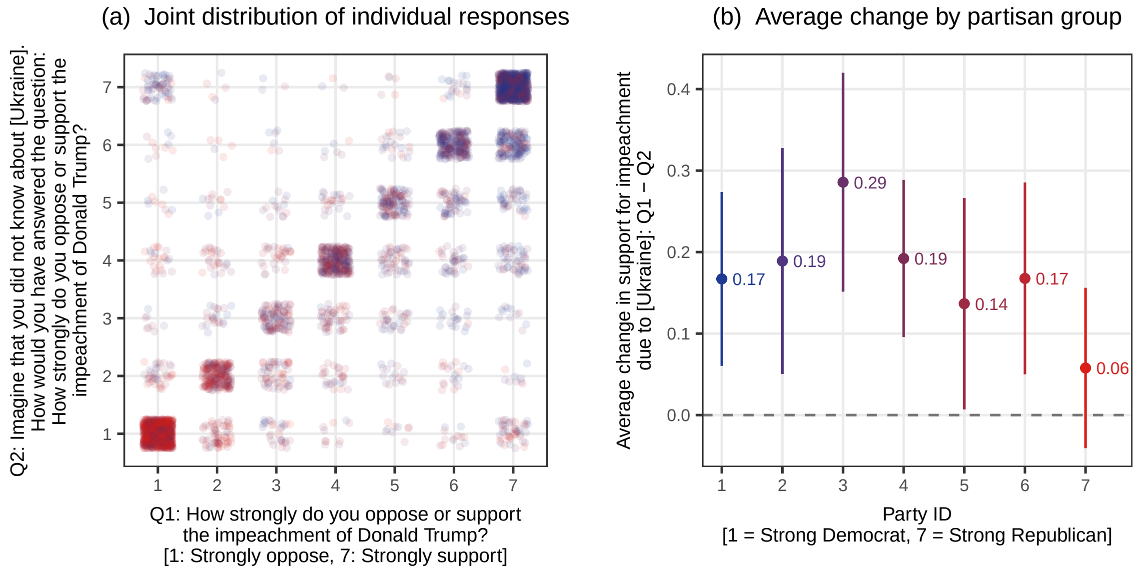

Figure 1: Figure 1. Counterfactual format example: the joint distribution of the two impeachment questions, and the average change by partisan group.

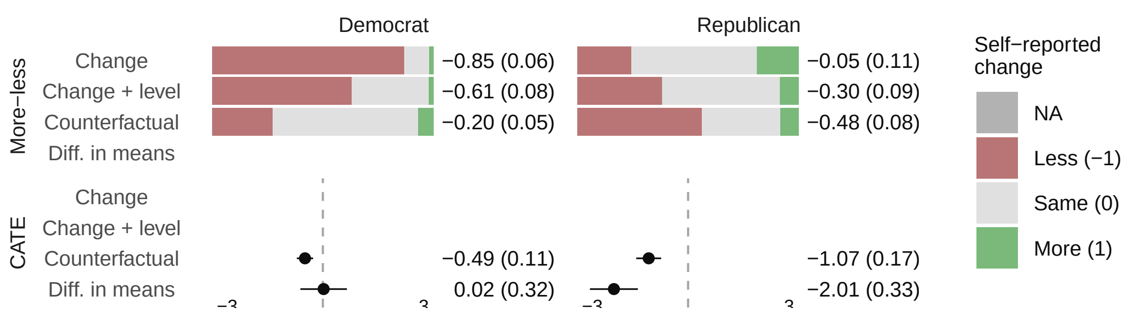

Figure 2: Figure 2. Self-reported change in the Cornish example, by party: three question formats against the experimental difference in means.

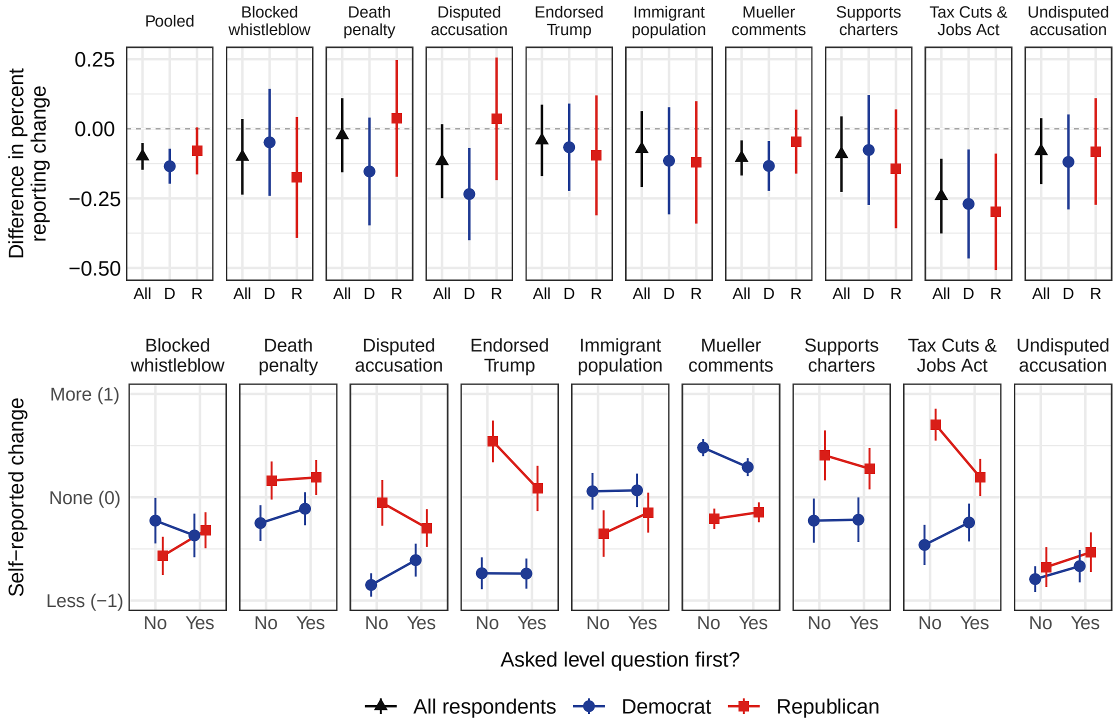

Figure 3: Figure 3. The effect of asking the level question first, on reporting any attitude change (a) and on the sign of self-reported change (b).

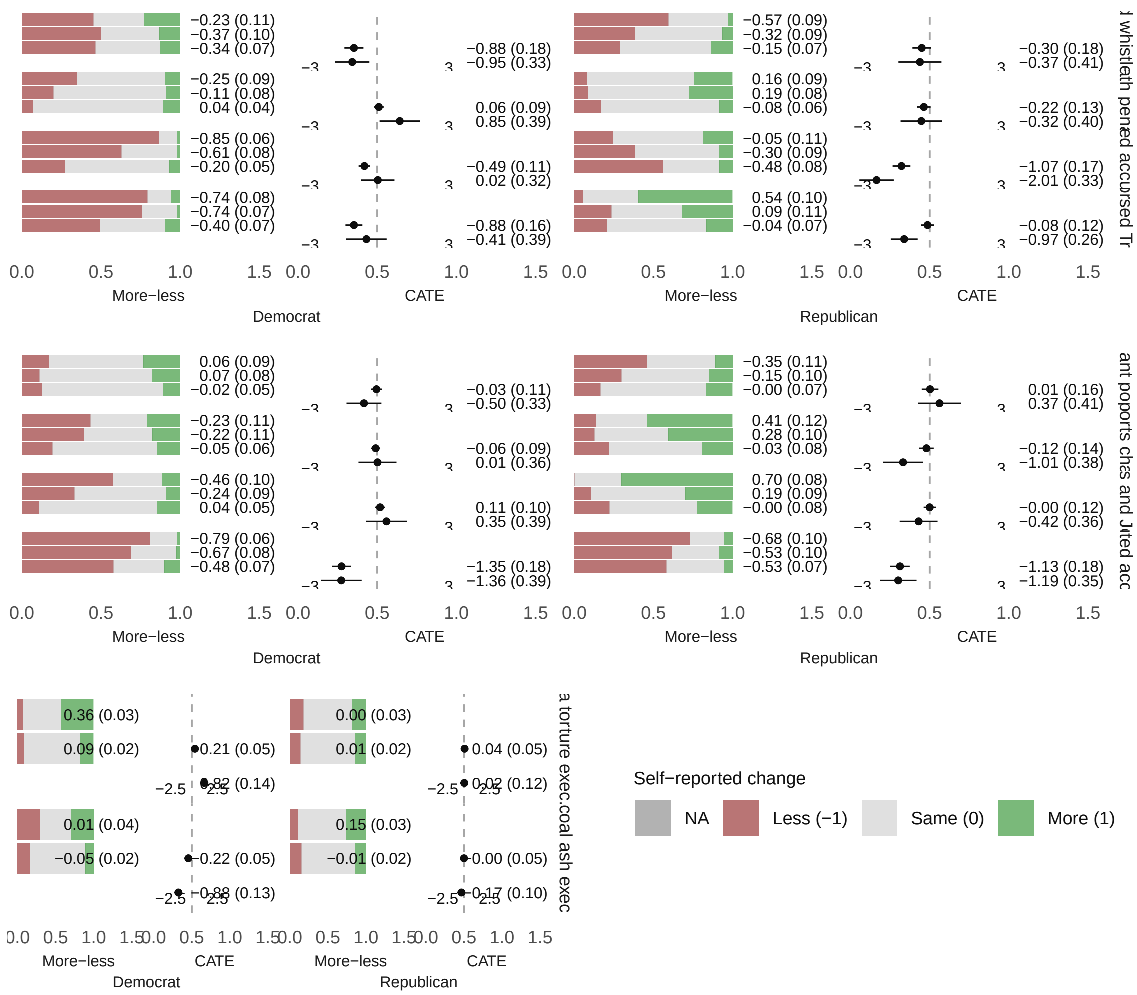

Figure 4: Figure 4. The full comparison of the change format, the randomized counterfactual format and the experiment, over the ten treatments in Studies 1 and 2a.

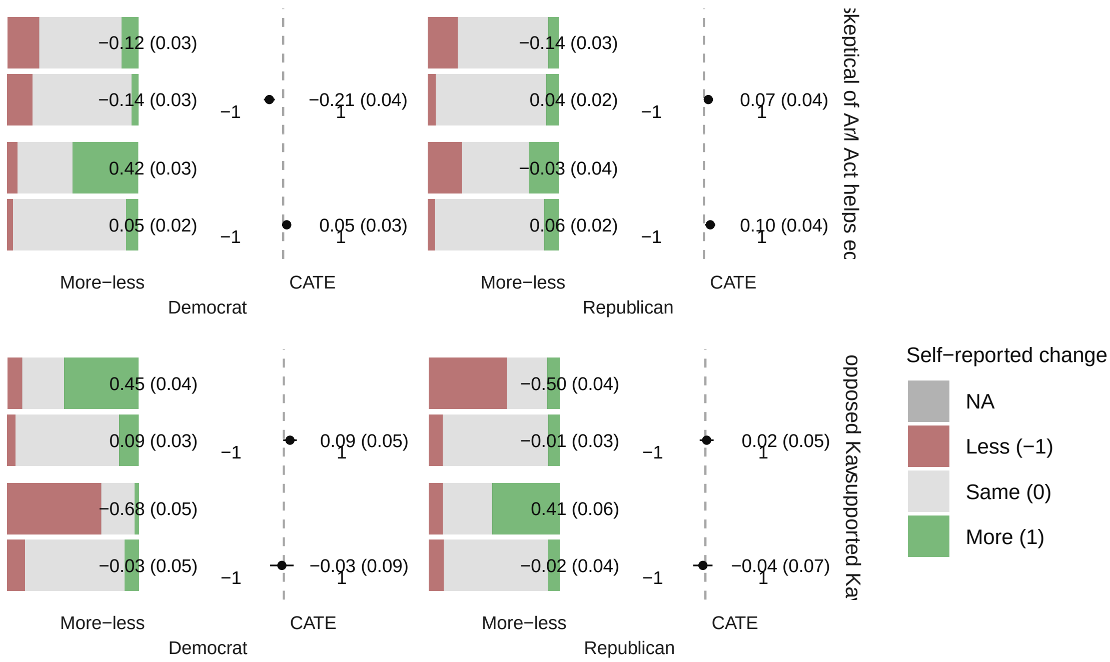

Figure 5: Figure 5. The change format against the nonrandomized counterfactual format, for the Study 2a treatments subjects may have met before the survey.

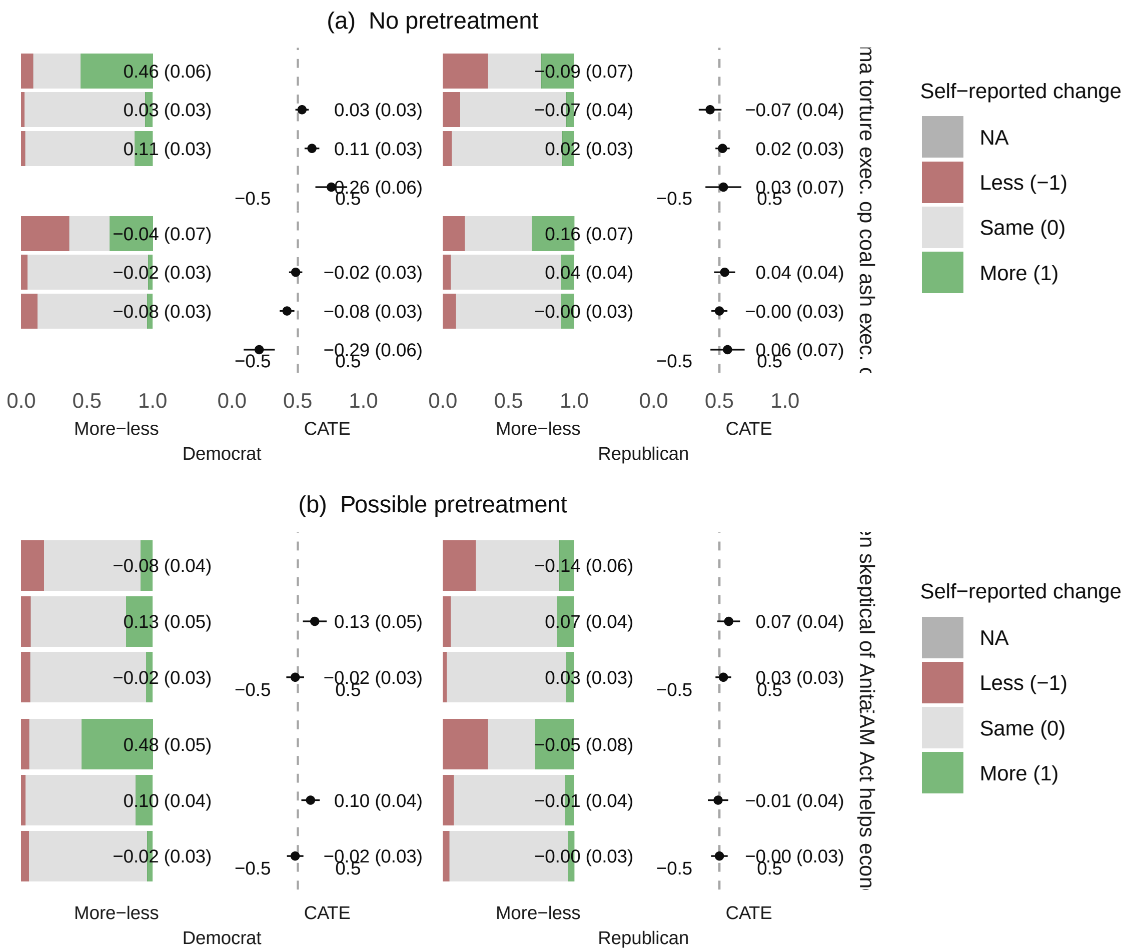

Figure 6: Figure E.2. Study 2b, setting the simultaneous outcomes format against the change and counterfactual formats.

7 Rewrite Verification

Table 8: Rewrite verification: 50 of 57 claims match the published values

Table or figure	Claim	Published	Rewrite	Match	Locus
Text, p. 31	Survey responses obtained from Lucid	4034.00	4034.0000	1	NA

Text, p. 32	Correlation of the two impeachment questions	0.82	0.8285	0	paper_internal
Text, p. 32	Share reporting exactly zero change	0.72	0.7209	1	NA
Text, p. 32	Average difference between the two questions	0.16	0.1601	1	NA
Text, p. 32	Standard error of that average	0.02	0.0226	1	NA
Text, p. 38	Cornish, Democrats reporting less likely (per cent)	87.00	86.7920	1	NA
Text, p. 38	Cornish, Republicans reporting no effect (per cent)	57.00	56.7570	1	NA
Text, p. 39	Cornish, effect of the level question on Democrats' more-minus-less score	0.24	0.2404	1	NA
Text, p. 39	Standard error of that effect	0.10	0.0972	1	NA
Text, p. 39	Cornish, same effect among Republicans	-0.24	-0.2438	1	NA
Text, p. 39	Standard error of that effect	0.14	0.1421	1	NA
Text, p. 40	Study 1 N	417.00	417.0000	1	NA
Text, p. 40	Study 2a N	2475.00	2475.0000	1	NA
Text, p. 44	Study 2b N	1110.00	1110.0000	1	NA
Text, p. 44	Study 3 N	1074.00	1023.0000	0	archive
Text, p. 44	Decrease in reporting any change, all respondents (points)	10.00	9.9271	1	NA
Text, p. 44	Standard error of that effect (points)	2.00	2.4431	1	NA
Text, p. 44	Same decrease among Democrats (points)	14.00	13.4590	0	paper_internal
Text, p. 44	Standard error, Democrats (points)	3.00	3.1890	1	NA
Text, p. 44	Same decrease among Republicans (points)	8.00	7.9455	1	NA
Text, p. 44	Standard error, Republicans (points)	4.00	4.2791	1	NA
Text, p. 44	Same decrease among pure independents (points)	2.00	2.2329	1	NA
Text, p. 44	Standard error, independents (points)	6.00	6.1777	1	NA
Text, p. 46	Cells compared, change format against the experiment	20.00	20.0000	1	NA
Text, p. 46	Change format has the opposite sign	12.00	12.0000	1	NA
Text, p. 46	Counterfactual format has the opposite sign	3.00	3.0000	1	NA

Text, p. 46	Opportunities to evaluate a counterfactual guess	40.00	40.0000	1	NA
Text, p. 47	Difference in means tests rejecting at $p < 0.05$	12.00	12.0000	1	NA
Text, p. 46	Comparisons of the experiment against the counterfactual guess	20.00	20.0000	1	NA
Text, p. 46	Differences whose bootstrap interval excludes zero	6.00	6.0000	1	NA
Figure 1b	Average change, party ID 1	0.17	0.1670	1	NA
Figure 1b	Average change, party ID 2	0.19	0.1890	1	NA
Figure 1b	Average change, party ID 3	0.29	0.2857	1	NA
Figure 1b	Average change, party ID 4	0.19	0.1921	1	NA
Figure 1b	Average change, party ID 5	0.14	0.1367	1	NA
Figure 1b	Average change, party ID 6	0.17	0.1677	1	NA
Figure 1b	Average change, party ID 7	0.06	0.0578	1	NA
Figure 3a	Estimates plotted (ten facets by All, Democrat, Republican)	30.00	30.0000	1	NA
Figure 3a	Pooled decrease in reporting any change, all respondents	0.10	0.0993	1	NA
Figure 3b	Means plotted (nine facets by two formats by two parties)	36.00	36.0000	1	NA
Figure 3b	Endorsed Trump, Republicans: sign of the level question's effect	-1.00	-1.0000	1	NA
Figure 3b	Mueller comments, Democrats: sign of the level question's effect	-1.00	-1.0000	1	NA
Figure 2	Printed labels agreeing with the published figure	10.00	10.0000	1	NA
Figure 4	Printed labels agreeing with the published figure	96.00	95.0000	0	environment
Figure 5	Printed labels agreeing with the published figure	24.00	24.0000	1	NA
Figure E.2a	Printed labels agreeing with the published figure	24.00	24.0000	1	NA
Figure E.2b	Printed labels agreeing with the published figure	20.00	20.0000	1	NA
Figure 4	Undisputed accusation, Republican, CATE, Counterfactual	-1.12	-1.1250	0	environment
Table D.3	Cells agreeing with the published table	336.00	336.0000	1	NA
Table D.3	Cells agreeing if the column headings are read as printed	336.00	172.0000	0	paper_internal

Table D.3	Sample sizes agreeing	84.00	84.0000	1	NA
Table D.4	Cells agreeing with the published table	216.00	216.0000	1	NA
Table D.4	p-values agreeing	72.00	72.0000	1	NA
Table D.4	Sample sizes agreeing	72.00	72.0000	1	NA
Table D.5	Point estimates agreeing with the published table	60.00	60.0000	1	NA
Table D.5	Sample sizes agreeing	60.00	60.0000	1	NA
Table D.5	Bootstrap standard errors and interval endpoints agreeing	60.00	53.0000	0	rewrite

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8 R Environment

Table 9: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
ggh4x	0.3.1
gridExtra	2.3
rsample	1.3.2
knitr	1.51
kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)